

CHAPTER 3
OCEANIC NEKTON

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

This chapter is divided into four parts: COMPOSITION OF THE OCEANIC NEKTON, ENVIRONMENTAL CONDITIONS, ADAPTATIONS OF OCEANIC NEKTON, and ECOLOGY OF NEKTON.

I. Composition of the Oceanic Nekton

Little is known about the ecology and life history of the nekton, but using information about their anatomy and physiology, we can infer a good deal about these organisms. What we are dealing with here is a major portion of the charismatic megafauna.

cetaceans	ichthyosaurs	pinnipeds
epipelagic	meroepipelagic	plesiosaurs
fishes	mosasaurs	sharks
holoepipelagic	nekton	

A variety of organisms compose the nekton, including bony fish, sharks, mammals, reptiles, and birds. A few invertebrates qualify as nekton (for example, Molluca and Cephalopoda). Fish that spend their entire lives as nekton in the epipelagic zone are called holoepipelagic (G., *holo*, whole). Those that spend only part of their life in the epipelagic are termed meroepipelagic (G., *mero*, a part). Midwater fish will be considered in Chapter 4 (DEEP-SEA BIOLOGY). While, perhaps in a rigorous sense, seabirds are not nekton, they do feed at the surface, and some may dive to depths approaching 100 m.

II. Environmental Conditions

While epipelagic waters are profoundly three dimensional, this condition does not mean that the environment is uniform. As described in Chapter 2 (PLANKTON AND PLANKTON COMMUNITIES), pelagic waters can be highly stratified. Nevertheless, there are no structures or substrata that may be used as shelter (refuge).

The concept of three dimensionality should not be dismissed as simple or trivial. For example, consider an animal with which we all are somewhat familiar: the ground squirrel. A squirrel's world is not a uniform three-dimensional universe. Basically, squirrels live in a world of surfaces. The ground is a large, flat—or, at best, rolling—surface. Tree trunks are long, curved surfaces, while branches are long, highly curved surfaces; the smallest branches are just lines. The behavior of these arboreal animals is very much influenced by their perceptions of the world; chase a squirrel around a tree (or back and forth to either side of the tree trunk) and you will see my meaning. To the squirrel, a clear and present danger is seen and then simply avoided by moving to one side or the other. If the danger reappears, the squirrel moves again. While squirrels are well adapted to their environment, roads present serious problems for them. In this case, the threat (a car) does not disappear when the squirrel moves to the right or back again to the left, as it might were this dance being done around a tree trunk. Thus, squirrels appear to line themselves up on your car's wheels as you approach. The lesson for us folk with limited appreciation of such three-dimensional worlds is not to jump so quickly to assume that we understand it all.

III. Adaptations of Oceanic Nekton

Upon reflection, we find that oceanic nektonic life possesses features that are very similar to aspects of terrestrial life, as well as features that are very different. This, the longest part of the chapter, is divided into nine sections:

- Buoyancy
- Locomotion
- Surface of Resistance and Body Shape
- Defense and Camouflage
- Sensory Systems
- Echolocation
- Reproduction and Life Cycles
- Migrations
- Special Adaptations of Marine Birds and Mammals

acoustic lens	frictional resistance	oxygen debt
altricial	gas bladder	panting
ampullae of Lorenzini	gas gland	physoclist
anaerobic glycolysis	geomagnetism	physostome
bends	harem	pinniped
blubber	heterocercal tail	polygamous
bradycardia	homocercal tail	precocial
buoyancy	hydrofoil	residual weight
camouflage	induced drag	rete mirabile
caudal peduncle	keel	rookery
celestial compass	lactic acid	salt gland
countercurrent system	laminar flow	sonar
countershading	lateral line	spermaceti organ sac
cryptic body shape	lift	streamline
cryptic coloration	lunate tail	swim bladder
deep scattering layer (DSL)	magnetite, Fe_3O_4	territorial behavior
dimethyl sulfide	mammalian dive response	thermoregulation
drag	map sense	transparency
echolocation	mass stranding	tube nosed
eddies	melon	turbulent flow
electroreception	migration	vortices
fermentation	myoglobin	
form resistance	oval	

1. Features that improve buoyancy, enhance streamlining, and increase speed are common adaptations of the oceanic nekton. Buoyancy can be achieved in a variety of ways, but most fish possess swim bladders of one of two types: physostome or physoclist. In the former, there is an open duct between the gas bladder and the esophagus, while in the latter, no such duct exists. Gas bladders cannot respond quickly to large changes in pressure; therefore, fast-swimming fish that range far in the water column tend to lack these structures. Other adaptations to increase buoyancy include replacement of heavy chemical ions by lighter ones, storage of fats and oils, storage of air amongst fur and feathers, and development of hydrodynamic lifting surfaces (for example, pectoral fins and flippers).
2. The undulatory motions of the body and fins of swimming nektonic fish are universally similar, differing mainly in the amplitude of the wave. Three types of resistance must be overcome by organisms moving through water: frictional resistance, form resistance, and induced drag.
3. Even large-bodied, fast-swimming nektonic organisms are somewhat camouflaged. While camouflage by transparency is not found in the nekton, alteration of body shape and cryptic coloration (countershading) do serve to reduce visibility of these organisms.

4. The sense organs of nektonic animals are well developed. Important sense organs include the lateral-line system, ampullae of Lorenzini, eyes, and ears. One sensory system about which we know little is geomagnetic sense. The sense of hearing is well developed in mammals, especially the cetaceans. This group is highly adapted for the production and reception of sound; some use their capabilities for echolocation. Echolocation is used by animals in a manner similar to the use of sonar. Low-frequency sound is used to locate objects, while high-frequency sound is used to provide finer detail about the objects.
5. The large fat-filled organ of the toothed whales, called the melon or spermaceti organ, gives these animals a bulging forehead. In sperm whales, this structure can run 40% of the animal's length. These organs and their elaborate associate structures produce and detect sound over a large frequency range.
6. Many holonektonic fish produce vast numbers of embryos (*r*-selection) that are cast to develop on their own accord and, therefore, are subject to massive losses due to predation. Pelagic sharks, on the other hand, produce few offspring (*K*-selection), endowing each with more energy. They are, of course, more well developed at birth, too.
7. Marine birds and turtles lay their eggs on land, but there is a great difference in the capabilities of the newly hatched young: Young birds are helpless (altricial), while young turtles are very active (precocial). Pinnipeds give birth to young that tend to be altricial; cetaceans have offspring that are precocial. Although they remain close to their mother after birth, young whales can swim immediately.
8. Many marine animals migrate. Important migratory species include many reptiles, birds, mammals, and fish. The mechanisms used by these animals to orient themselves are as remarkable as the distances they migrate. The presence of membrane-bound magnetite in tuna suggests an anatomical basis for magnetotaxis. An explanation for the phenomenon of the mass stranding of whales has not been confirmed, but it may be related to magnetotaxis.
9. Marine endotherms possess special adaptations permitting life in the ocean. These features include reduced surface-to-volume ratio, insulating fat, feathers and fur, and countercurrent heat-exchange circulation in their extremities.

Diving mammals possess a complex physiological process known as the dive response that permits these animals to hold their breath for a long time while being active underwater. The dive response is not unique to diving mammals. Humans possess it, although to a much lesser degree, but in rabbits the response is profound and is triggered by the inhalation of smoke. A brief explanation of the dive response in marine mammals, followed by a few statements about this phenomenon in rabbits, could be an excellent way to generate a good classroom discussion. The concept of a bunny with scuba gear (that is, a diving rabbit) may bring more than a few laughs. It's a lesson that I find that the students do not forget.

Why would it be adaptive for rabbits at home in their warren to possess such a curious physiological phenomenon? Discuss and compare this concept to the dive response in marine mammals.

Evans, D.H. (ed.). 1993. *The physiology of fishes*. Boca Raton, FL: CRC Press, Inc.

IV. Ecology of Nekton

The trophic ecology of most adult nekton may be summed up in one word: carnivory. However, the methods of feeding used by these animals vary widely. Some of the animals are voracious raptors (for example, sperm whales and salmon), while others filter feed (for example, baleen whales). Two sections compose this part of the chapter:

- Feeding Ecology and Food Webs
- Ecological Significance of Marine Mammals

<i>Architeuthis</i>	euphausiids	tuna
baleen	krill	whalebone
<i>Euphasia superba</i>	rorqual whales	

1. Most nekton feed on other nekton. Plankton feeders range from the smallest fish (sardines) to the large baleen whales.
2. Not surprisingly, food webs differ in different parts of the ocean. In the colder part of the ocean, diatoms occupy the lowest trophic level. However, in tropical waters, dinoflagellates and coccolithophores are the primary producers.
3. A complete understanding of the ecological energetics of the world's cetacean population has not been accomplished. Evidence suggests that cetaceans may be extremely important, perhaps composing over two times the current worldwide fisheries catch. With the reduced numbers of whales in the Antarctic waters ($\leq 90\%$ reduction), birds and pinnipeds have increased in number by several-fold. Whales also are important when they die and their carcasses sink to the bottom. There, they provide a bountiful supply of energy-rich protein and fat to the benthic community. Gray whales also influence the environment by disturbing the benthos while feeding. Other marine mammals influence their environment in less well-recognized ways, as documented in the text.

ENRICHMENT EXERCISES

Siren, incertae sedis

Little is known about mermaids (Nixi, Sirenia), but they are more than a historical curiosity. Banse (1990) has provided the most thorough narrative on these enigmatic creatures. Banse's work rivals that of previous workers who have investigated the Loch Ness Monster (a.k.a. Nessie). Some references on these two phenomena are as follows:

Banse, K. 1990. Mermaids—Their biology, culture, and demise. *Limnol. Oceanogr.* 35: 148–153.

Dahlerup, P. 1990. The little mermaid deconstructed. *Scandinavian Studies* 62: 418–428.

Scheider, W., and P. Wallis. 1973. An alternative method of calculating the population density of monsters in Loch Ness. *Limnol. Oceanogr.* 18: 343.

Sheldon, R.W., and S.R. Kerr. 1972. The population density of monsters in Loch Ness. *Limnol. Oceanogr.* 17: 796–798.

Sheldon, R.W., and S.R. Kerr. 1973. The Loch Ness monster: reply to the comments of C.H. Mortimer. *Limnol. Oceanogr.* 18: 345–346.

For those who are provoked to anger by this whimsy, I tender Puck's Epilogue:

“If we shadows have offended, think but this, and all is mended;
That you have but slumber'd here while these visions did appear.”
—Shakespeare, *A Midsummers Night's Dream*, 5:1:430–433

Oceanic Nekton in the Cretaceous Period

Talking about Nessie may not be the only way to stimulate class discussion of oceanic nekton. Having the students discuss what the ocean food webs might have been in the Cretaceous period would be a good way to make sure that they appreciate and have mastered the wealth of information discussed in this chapter. Again, I apologize should anyone consider this topic to be just complete whimsy. Some useful references are as follows:

Bakker, R. 1993. Jurassic sea monsters. *Discover* 14: 78–85.

Cruickshank, A.R.I. 1991. Dorsal nostrils and hydrodynamically driven underwater olfaction in plesiosaurs. *Nature* 352: 62–64.

Halstead, L.B. 1989. Plesiosaur locomotion. *J. Geol. Soc.* 146: 37–41.

Humphries, S., and G.D. Ruxton. 2002. Why did some ichthyosaurs have such large eyes? *J. Exp. Bio.* 205: 439–441.

Motani, R. 1996. Eel-like swimming in the earliest ichthyosaurs. *Nature* 382: 347–348.

- Motani, R. 1999. Large eyeballs in diving ichthyosaurs. *Nature* 402: 747.
- Motani, R. 2000. Rulers of the Jurassic seas. *Sci. Am.* 2000 283(6): 52–60.
- Sato, T. 1998. Cretaceous plesiosaurs ate ammonites. *Nature* 394(6694): 629–630.
- Teresi, D. 1998. Monster of the tub. *Discover* 19(4): 86–92.

Marine Mammals and ATOC

Acoustic thermography of ocean climate (ATOC) is controversial because of the magnitude of the sound energy used and the possibility that this sound could interfere with or even damage the hearing of cetaceans. You may wish to engage your students in a discussion on the issues generated by both sides of the debate. Following are a few useful references to get you started:

- Anderson, R.B. 1991. The heard shot round the world on marine mammals. *Natural History* (March): 76.
- Au, W.W.L., P.E. Nachtigall, and J.L. Pawloski. 1997. Acoustic effects of the ATOC signal (75 Hz, 195 dB) on dolphins and whales. *J. Acoustical Soc. Am.* 101: 2973–2977.
- Boehlert, G.W., et al. 2001. Autonomous pinniped environmental samplers: Using instrumented animals as oceanographic data collectors. *J. Atmos. Ocean. Tech.* 18: 1882–1893.
- Buckingham, M.J., J.R. Potter, and C.L. Epifanio. 1996. Seeing underwater with background noise. *Sci. Am.* 274(2): 86–90.
- Costa, D.P., and S.A. Hayes. 2001. Underwater sound and marine mammals. Pages 405–407 in *Yearbook of science and technology*. New York: McGraw Hill.
- Frankel, A.S., and C.W. Clark. 2000. Behavioral responses of humpback whales (*Megaptera novaeangliae*) to full-scale ATOC signals. *J. Acoust. Soc. Am.* 108: 1930–1937.
- Frankel, A.S., and C.W. Clark. 2002. ATOC and other factors affecting the distribution and abundance of humpback whales (*Megaptera novaeangliae*) off the north shore of Kauai. *Mar. Mam. Sci.* 18: 644–662.
- Kumagai, J. 1994. Concern for marine mammals delays oceanic temperature-taking. *Physics Today* 47: 85–86.
- Malanotte-Rizzoli, P. (ed.). 1996. *Modern approaches to data assimilation on ocean modeling*. Amsterdam: Elsevier Science BV.
- Munk, W., P. Worcester, and C. Wunsch. 1995. *Ocean acoustic tomography*. New York: Cambridge University Press.
- Nicklin, F. 1995. Bowhead whales. *Nat. Geo.* 188(2): 114–129.
- Schmidt, K. 1994. ATOC delayed as report laments research gaps. *Science* 264: 339–340.
- Shurkin, J. 1994. Pacific sound experiment faces heat. *Nature* 368: 485.
- Spindel, R.C. 1990. Ocean acoustic tomography. *Sci. Am.* 263(4): 94–99.
- Whitehead, H. 1995. Sperm whales. *Nat. Geo.* 188(5): 57–73.
- See also the Marine Mammal Protection Act (MMPA) of 1972:
http://www.nmfs.noaa.gov/prot_res/laws/MMPA/MMPA.html.

Undersea Flight

A good activity that could be employed as a cooperative-learning opportunity is to have your students view a video that shows penguins (*Pygoscelis papua*) swimming underwater and then to have them watch a slow-motion video of the standard flight of birds (that is, not hovering). Have the students discuss the following question: Do penguins paddle through the water, or do they “fly” through it like birds? Your students must understand the basic principles of bird flight for this discussion to be meaningful.

